(2) Min-Max Normalization (Rescaling)

→ Min-max normalization rescales data into a fixed range usually [0-1] or [-1,1]

$$x' = \frac{x - min(x)}{max(x) - min(x)} \Rightarrow \text{for range}(0,1)$$

→ To scale feature into the same fixed range so that no feature dominates

→ best when we need values bounded.

→ often used in ML, DL where input value should not grow too large.

→ if features have diffent ranges (salary in lakhs vs age in years), large scale feature dominates and bias the model.

. min-max ensures all features contribute equally by scaling to [0,1] or [-1,1].

Exp

Age = [20-60]

Salary = [20,000 - 100000]

without normalization: Salary dominates Age in distance calculation.

with min-max both age and salary get scaled into 0-1 for contributions in ML.

Exp

score [10, 12, 12, 13, 15, 18, 20, 22]

n = 8

min-max normalization.

min = 10, max = 22, range = 22-10

= 12

step ② scale to [0,1] = $x' = \frac{x - min}{max - min} \Rightarrow \frac{x-10}{12}$

Contribute

17.

10 → 0/12 ⇒ 0.000

12 → 2/12 ⇒ 0.1667

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13 → 3/12 ⇒ 0.2500

15 → 5/12 ⇒ 0.4167

18 → 8/12 ⇒ 0.6667

20 → 10/12 ⇒ 0.8333

22 → 12/12 ⇒ 1.0000

Generic formula for Any custom range

$[a, b]$

You can generalize the formula to scale your data to any arbitrary range $[a, b]$

e.g. $[-1, 1]$ or $[0, 100]$

$$X_{normalized} = a + \frac{((X - X_{min}) * (b - a))}{(X_{max} - X_{min})}$$

where.

→ a is the new minimum of your desired range

→ b is the new maximum of your desired

How it works:-

1. $(X - X_{min}) / (X_{max} - X_{min})$ first scales the value to the range $[0,1]$.

2. multiply by $(b-a)$ scale it to the range $[0,(b-a)]$

Adding a then shift the range to the final desired range. $[a,b]$

Desired

Ex dataset of Ages [25, 36, 19, 44, 50]

range [0,1].

1. Find min and max

$X_{min} = 19$

$X_{max} = 50$

2. Apply the formula for each value

[0.194, 0.548, 0.0, 0.8060, 1.0]

Ex.

(2) a custom range [-1, 1]

dataset [25, 36, 19, 44, 50]

$a = -1, \quad b = 1$

$X_{min} = 19, \quad X_{max} = 50$

[-0.613, 0.097, -1.0, 0.613, 1.0].